

```
!pip install -q "crewai==0.28.8" "crewai-tools==0.1.6" \
  "langchain==0.1.20" "langchain-core==0.1.53" \
  "langchain-community==0.0.38" "langchain-openai==0.1.7"
```

Show hidden output

```
!pip install -q "crewai==0.28.8" "langchain==0.1.20" "langchain-core==0.1.53" "langchain-community==0.0.38"
```

Show hidden output

```
!pip install -q -U crewai crewai-tools
```

Show hidden output

```
!pip install -q "numpy==1.26.4"
```

```
# Warning control
import warnings
warnings.filterwarnings('ignore')
```

```
from crewai import Agent, Task, Crew
```

```
# import requests
# key = "your-api-key"
# r = requests.post(
#     f"https://generativelanguage.googleapis.com/v1beta/models/gemini-2.0-flash:generateContent?key={key}",
#     json={"contents": [{"parts": [{"text": "say hi"}]}]},
# )
# print(r.status_code)
# print(r.text[:300])
```

```
import os
# os.environ["OPENAI_API_KEY"] = "your-openAI-key"
# os.environ["OPENAI_MODEL_NAME"] = "gpt-3.5-turbo"
# os.environ["GEMINI_API_KEY"] = "your-gemini-key"

os.environ["Deep_API_KEY"] = "your-deepseek-key"
```

```
os.environ["SERPER_API_KEY"] = "your-serper-key"
```

```
from langchain_openai import ChatOpenAI

# llm = ChatOpenAI(
#     model="gemini-2.0-flash",
#     base_url="https://generativelanguage.googleapis.com/v1beta/openai/",
#     api_key=os.environ["GEMINI_API_KEY"],
#     temperature=0.3,
# )

llm = ChatOpenAI(
    model="deepseek-chat",
    base_url="https://api.deepseek.com/v1",
    api_key=os.environ["Deep_API_KEY"],
    temperature=0.7,
)
```

```
from crewai_tools import ScrapeWebsiteTool, SerperDevTool
```

```
search_tool = SerperDevTool()
scrape_tool = ScrapeWebsiteTool()
```

Data analyst agent to analyze market to find us good opportunities to invest

```
data_analyst_agent = Agent(
    role="Data Analyst",
    goal="Monitor and analyze market data in real-time "
    "to identify trends and predict market movements.",
    backstory="Specializing in financial markets, this agent "
    "uses statistical modeling and machine learning "
    "to provide crucial insights. With a knack for data, "
    "the Data Analyst Agent is the cornerstone for "
    "informing trading decisions.",
```

```

        verbose=True,
        llm=llm,
        allow_delegation=True,
        tools = [scrape_tool, search_tool]
    )

```

look at options and weigh in what are the movements that we should do regarding any stocks

```

trading_strategy_agent = Agent(
    role="Trading Strategy Developer",
    goal="Develop and test various trading strategies based "
        "on insights from the Data Analyst Agent.",
    backstory="Equipped with a deep understanding of financial "
        "markets and quantitative analysis, this agent "
        "devises and refines trading strategies. It evaluates "
        "the performance of different approaches to determine "
        "the most profitable and risk-averse options.",
    verbose=True,
    llm=llm,
    allow_delegation=True,
    tools = [scrape_tool, search_tool]
)

```

This agent will look at the timing, price, figure out entry point

```

execution_agent = Agent(
    role="Trade Advisor",
    goal="Suggest optimal trade execution strategies "
        "based on approved trading strategies.",
    backstory="This agent specializes in analyzing the timing, price, "
        "and logistical details of potential trades. By evaluating "
        "these factors, it provides well-founded suggestions for "
        "when and how trades should be executed to maximize "
        "efficiency and adherence to strategy.",
    verbose=True,
    llm=llm,
    allow_delegation=True,
    tools = [scrape_tool, search_tool]
)

```

Risk of buying stock and balancing things out

```

risk_management_agent = Agent(
    role="Risk Advisor",
    goal="Evaluate and provide insights on the risks "
        "associated with potential trading activities.",
    backstory="Armed with a deep understanding of risk assessment models "
        "and market dynamics, this agent scrutinizes the potential "
        "risks of proposed trades. It offers a detailed analysis of "
        "risk exposure and suggests safeguards to ensure that "
        "trading activities align with the firm's risk tolerance.",
    verbose=True,
    llm=llm,
    allow_delegation=True,
    tools = [scrape_tool, search_tool]
)

```

Creating tasks

Task 1: Data analysis: Monitoring the market, trying to find good stocks

```

# Task for Data Analyst Agent: Analyze Market Data
data_analysis_task = Task(
    description=(
        "Continuously monitor and analyze market data for "
        "the selected stock ({stock_selection}). "
        "Use statistical modeling and machine learning to "
        "identify trends and predict market movements."
    ),
    expected_output=(
        "Insights and alerts about significant market "
        "opportunities or threats for {stock_selection}."
    ),
    agent=data_analyst_agent,
)

```

Strategy

```
# Task for Trading Strategy Agent: Develop Trading Strategies
strategy_development_task = Task(
    description=(
        "Develop and refine trading strategies based on "
        "the insights from the Data Analyst and "
        "user-defined risk tolerance ({risk_tolerance}). "
        "Consider trading preferences ({trading_strategy_preference})."
    ),
    expected_output=(
        "A set of potential trading strategies for {stock_selection} "
        "that align with the user's risk tolerance."
    ),
    agent=trading_strategy_agent,
)
```

Plan trade execution

```
# Task for Trade Advisor Agent: Plan Trade Execution
execution_planning_task = Task(
    description=(
        "Analyze approved trading strategies to determine the "
        "best execution methods for {stock_selection}, "
        "considering current market conditions and optimal pricing."
    ),
    expected_output=(
        "Detailed execution plans suggesting how and when to "
        "execute trades for {stock_selection}."
    ),
    agent=execution_agent,
)
```

Risk assessment

```
# Task for Risk Advisor Agent: Assess Trading Risks
risk_assessment_task = Task(
    description=(
        "Evaluate the risks associated with the proposed trading "
        "strategies and execution plans for {stock_selection}. "
        "Provide a detailed analysis of potential risks "
        "and suggest mitigation strategies."
    ),
    expected_output=(
        "A comprehensive risk analysis report detailing potential "
        "risks and mitigation recommendations for {stock_selection}."
    ),
    agent=risk_management_agent,
)
```

```
!pip install -q "langchain-openai==0.1.7"
```

Creating the crew

The Process class helps to delegate the workflow to the Agents (kind of like a Manager at work) In the example below, it will run this hierarchically. manager_llm lets you choose the "manager" LLM you want to use.

```
from crewai import Crew, Process
from langchain_openai import ChatOpenAI

# Define the crew with agents and tasks
financial_trading_crew = Crew(
    agents=[data_analyst_agent,
            trading_strategy_agent,
            execution_agent,
            risk_management_agent],

    tasks=[data_analysis_task,
            strategy_development_task,
            execution_planning_task,
            risk_assessment_task],

    # manager_llm=llm,
```

```
process=Process.sequential,  
max_rpm=10,  
verbose=True  
)
```

WARNING:opentelemetry.trace:Overriding of current TracerProvider is not allowed

Running the Crew Set the inputs for the execution of the crew.

```
# Example data for kicking off the process  
financial_trading_inputs = {  
    'stock_selection': 'AAPL',  
    'initial_capital': '100000',  
    'risk_tolerance': 'Medium',  
    'trading_strategy_preference': 'Day Trading',  
    'news_impact_consideration': True  
}
```

```
### this execution will take some time to run  
result = financial_trading_crew.kickoff(inputs=financial_trading_inputs)
```

```
| RSI Divergence | The RSI may show oversold or overbought conditions, but the price could continue moving in the same  
| Whipsaw Risk | In a volatile market, price may touch $291 or $300.77 multiple times, triggering several entries and  
| Timing Risk | The RSI conditions may be met at an inopportune time, such as just before a major news release (e.g.,
```

Mitigation Recommendations

1. **Add a Volume Filter for Short Trades:** For the overbought reversal setup, require that volume is not only below 80M
2. **Use a Wider Stop-Loss for Long Trades at \$291:** Place the stop-loss at \$287.50 (below the 52-week moving average or
3. **Limit to One Trade per Day:** To avoid whipsaw losses, take only one range trade per day. If the first trade is stop
4. **Avoid Trading During News Events:** Check the economic calendar for any scheduled news releases (e.g., Fed speeches,
5. **Implement a Time-Based Exit:** If a range trade is not profitable within 2 hours, close the position manually. Range

Strategy 3: Gap-Fill (Least Recommended)

Concept: Trades the tendency of price to fill gaps created by overnight or pre-market moves.

Risk Assessment

Risk Factor	Description	Severity	Probability
---	---	---	---
No Gap Present	As of the current context, no gap exists. The strategy may not be actionable, leading to missed opp		
Partial Gap Fill	Price may only partially fill the gap before reversing, resulting in a losing trade if the take-p		
Gap-and-Go	The gap may not fill at all if the momentum is too strong (e.g., a gap up on positive news that continu		
RSI Confirmation Failure	The strategy requires RSI oversold (<15) for gap-down fills or overbought (>85) for gap-u		
Overnight Gap Risk	If a gap occurs, it is often driven by significant news (e.g., earnings, macroeconomic events).		

Mitigation Recommendations

1. **Only Trade Gaps > \$2.00:** Small gaps (<\$2.00) are more likely to fill quickly but offer poor risk/reward. Focus on
2. **Use a 1:1 Risk/Reward Ratio:** For gap-fill trades, set the take-profit target at the gap fill level (previous day's
3. **Combine with Volume Analysis:** If a gap occurs, check the pre-market volume. If volume is exceptionally high (e.g.,
4. **Set a Time Limit:** If the gap is not filled within 3 hours of the open, close the trade. Gaps that do not fill quic
5. **Do Not Force the Trade:** If no gap occurs, do not take any trade under this strategy. It is better to wait for the

Overall Portfolio & Execution Risks

Risk Factor	Description	Severity	Probability
---	---	---	---
Overtrading	The execution plan suggests monitoring multiple levels and setups simultaneously. This can lead to ove		
Emotional Trading	After a loss, traders may deviate from the plan (e.g., moving stop-losses, increasing position s		
Capital Allocation Risk	The plan does not specify position sizing. Using too large a position on any single trade		
Technology Risk	Day trading relies on stable internet connection, fast execution, and reliable platform. A technic		
Market Liquidity Risk	Although AAPL is highly liquid, during fast-moving markets (e.g., news-driven selloffs), sli		

Mitigation Recommendations

1. **Implement Strict Position Sizing:** Risk no more than 1% of total trading capital per trade. For example, if capital
2. **Set a Daily Loss Limit:** Stop trading for the day if total losses reach 2% of capital (e.g., \$2,000 on \$100,000 cap
3. **Use Limit Orders for Entry:** Always use limit orders to enter trades at the specified price. Avoid market orders, w
4. **Have a Backup Plan:** Ensure that stop-losses are set as stop-market orders (not stop-limit) to guarantee execution.
5. **Review Trades Daily:** At the end of each trading day, review all trades to identify any deviations from the plan an

Final Risk Rating & Recommendation

```
from IPython.display import Markdown  
Markdown(result)
```


Comprehensive Risk Analysis Report for AAPL Day Trading Strategies (June 19, 2026)

Prepared by: Risk Advisor
Date: June 19, 2026
Subject: Risk Evaluation of Proposed Day Trading Strategies for Apple Inc. (AAPL)

Executive Summary

This report provides a detailed risk assessment of the proposed day trading strategies for AAPL (Apple Inc.) based on the market context as of June 18, 2026, with a closing price of 298.01 and after — hours price of 297.20. The analysis incorporates recent fundamental news (price hikes due to memory chip shortages, foldable iPhone delays), technical levels, volume conditions, and machine learning forecasts. The overall risk profile for the day is elevated due to a mixed-to-bearish sentiment overhang, increased volatility from fundamental catalysts, and the potential for false breakouts. The following sections break down the risks for each strategy and provide actionable mitigation recommendations.

Strategy 1: Two-Day High/Low Breakout (Most Recommended)

Concept: Capitalizes on breakouts from established two-day high (300.80) and low (296.50) levels with volume confirmation.

Risk Assessment

Risk Factor	Description
False Bullish Breakout	300.80 on low volume or momentary buying pressure, then reverses sharply due to the fundamental overhang (price hikes) * Bearish Breakout Trap * * A break below 296.50 could be a green flag
Stop-Loss Hunting	Large institutional players may push price just above 300.80 or just below 296.50 to trigger stop orders.
Volume Confirmation Failure	The strategy relies on volume exceeding the 50-day moving average. Low volume may indicate a lack of conviction.
Gap Risk	Overnight news (e.g., further details on price hikes) could cause a significant gap up or down.

Mitigation Recommendations

- Require Multi-Candle Confirmation:** Do not enter on the first 5-minute candle close above 300.80 or below 296.50. Wait for at least two consecutive 5-minute candles to close beyond the level with increasing volume. This reduces false breakouts.
- Tighten Stop-Loss on Bearish Breakout:** For the bearish setup, consider a tighter stop-loss at 298.00 (just above the previous day's close) instead of 299.50, as the negative sentiment makes a large upward reversal less likely. This reduces risk from 3.00 to 1.50.
- Use a Trailing Stop for Bullish Breakout:** If a bullish breakout occurs, do not rely solely on fixed take-profit targets. Use a trailing stop of \$1.50 from the highest price achieved after entry to capture additional gains if momentum continues, while protecting against sudden reversals.
- Avoid Trading in the First 15 Minutes:** The first 15 minutes after market open are highly volatile and prone to noise. Wait for the 9:45 AM ET 5-minute candle to confirm the breakout.
- Set a Daily Loss Limit:** If either the bullish or bearish breakout trade results in a loss, stop trading for the day. Do not attempt to "revenge trade" by taking the opposite breakout setup.

Strategy 2: Trading Ranges with RSI (Second Choice)

Concept: Profits from price oscillations within a defined range (291–303) using RSI extremes.

Risk Assessment

Risk Factor	Description
Support Breakdown	The fundamental headwinds (price hikes, delays) could cause price to break below the 291 support zone decisively, turning a long-term uptrend into a correction. * The RSI may show overbought conditions, but the price could continue moving in the same direction (e.g., RSI 70+). * Whipsaw Risk * * In a volatile market, price can move quickly between range highs and lows.
Timing Risk	The RSI conditions may be met at an inopportune time, such as just before a major news release.

Mitigation Recommendations

- Add a Volume Filter for Short Trades:** For the overbought reversal setup, require that volume is not only below 80M but also declining compared to the previous 5-minute candle. This indicates waning buying interest and increases the probability of a reversal.
- Use a Wider Stop-Loss for Long Trades at 291 :** * * Place the stop – loss at 287.50 (below the 52-week moving average or the next significant support level) instead of 289.00. This gives the trade more room to breathe if the support is tested but holds. The increased risk is justified by the potential reward (\$12).
- Limit to One Trade per Day:** To avoid whipsaw losses, take only one range trade per day. If the first trade is stopped out, do not re-enter the same setup. This prevents cumulative small losses from eroding capital.
- Avoid Trading During News Events:** Check the economic calendar for any scheduled news releases (e.g., Fed speeches, economic data) during the trading session. If a release is imminent, avoid entering new range trades until the market has absorbed the news.
- Implement a Time-Based Exit:** If a range trade is not profitable within 2 hours, close the position manually. Range-bound conditions can break down suddenly, and holding a losing position for too long increases risk.

Strategy 3: Gap-Fill (Least Recommended)

Concept: Trades the tendency of price to fill gaps created by overnight or pre-market moves.

Risk Assessment

Risk Factor	Description	Severity	Probability
No Gap Present	As of the current context, no gap exists. The strategy may not be actionable, leading to missed opportunities or forcing trades that do not align with the setup.	Low	High
Partial Gap Fill	Price may only partially fill the gap before reversing, resulting in a losing trade if the take-profit target is set too far away.	Medium	Medium
Gap-and-Go	The gap may not fill at all if the momentum is too strong (e.g., a gap up on positive news that continues higher). This is a classic failure mode of gap-fill strategies.	High	Medium
RSI Confirmation Failure	The strategy requires RSI oversold (<15) for gap-down fills or overbought (>85) for gap-up fills. These extreme RSI levels are rare and may not coincide with the gap fill level, leading to missed trades.	Medium	Medium
Overnight Gap Risk	If a gap occurs, it is often driven by significant news (e.g., earnings, macroeconomic events). The same news could cause the gap to widen further, making the fill less likely and increasing the risk of a losing trade.	High	Low

Mitigation Recommendations